

week-4-lesson-1

Task 1.

One large chemical company pays its salespeople on a commission basis. The salespeople receive \$200 per week plus 9% of their gross sales for that week. For example, a sales-person who sells \$5000 worth of chemicals in a week receives \$200 plus 9% of \$5000, or a total of \$650. Develop a program that will input each salesperson's gross sales for last week and will calculate and display that salesperson's earnings. Process one salesperson's figures at a time. Here is a sample input/output dialog:

```
Enter sales in dollars (-1 to end): 5000.00
```

```
Salary is: $650.00
```

```
Enter sales in dollars (-1 to end): 1234.56
```

```
Salary is: $311.11
```

```
Enter sales in dollars (-1 to end): 1088.89
```

```
Salary is: $298.00
```

```
Enter sales in dollars (-1 to end): -1
```

Task 2. 2D Coordinate Space Classifier (if-else & Compound Relationals)

Write a C program that prompts the user to enter a 2D coordinate point (x, y) using two float variables. The program must classify the exact location of the point in a coordinate grid and print its location.

- **Origin Identification:** The origin must be detected when both x and y are exactly 0.0.
- **Axis Detection:** The program must identify if the point lies exactly on the X-axis (y is zero, but x is not) or exactly on the Y-axis (x is zero, but y is not).
- **Quadrant Classification:** Categorize points lying within Quadrants I, II, III, or IV.
- **Syntax Constraint:** Ensure only one classification message is printed.

Task 3. Write a program that reads a nonnegative integer and computes and prints its factorial.

Task 4. A palindrome is a number or a text phrase that reads the same backward as forward. For example, each of the following five-digit integers is a palindrome: 12321, 55555, 45554 and 11611. Write a program that reads in a five-digit integer and determines whether or not it is a palindrome. [Hint: Use the division and remainder operators to separate the number into its individual digits.]

Task 5. Input an integer containing only 0s and 1s (i.e., a “binary” integer) and print its decimal equivalent. 234 can be interpreted as $4 * 1 + 3 * 10 + 2 * 100$. The decimal equivalent of binary 1101 is $1 * 1 + 0 * 2 + 1 * 4 + 1 * 8$ or $1 + 0 + 4 + 8$ or 13.

Task 6. Write for statements that print the following sequences of values:

1, 2, 3, 4, 5, 6, 7

3, 8, 13, 18, 23

20, 14, 8, 2, -4, -10

19, 27, 35, 43, 51

Task 7. The prices of coffee: large - 1000 KZT, medium - 800 KZT, small - 600 KZT. The program prompts the user to enter the amount three times for large, medium and small respectively and shows the total sum to be paid. Another way is the user enters L, M or S for the sizes respectively and then the number for the amount. Then the program shows the price to be paid.

Task 8. A user enters the number from 1 to 10 and the program prints the numbers starting from the entered number till 10. If the entered number is smaller than 4, the program skips number 4. If a user enters a number larger than 10 the program ends with an error message.

Task 9

Write a program that counts the number of digits in an integer using a do while loop. Ensure that negative value is correctly evaluated without using an if statement.